

Program 1:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main(){
    pid_t pid = fork();

    if(pid == 0){
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        sleep(10);
    }else{
        printf("Parent process\n");
        printf("PID : %d\n", getpid());

        wait(NULL);
    }
    return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64: ~/t  ×  +  v
crazyguy64@CrazyGuy64:~$ cd practical_3
crazyguy64@CrazyGuy64:~/practical_3$ nano program1.c
crazyguy64@CrazyGuy64:~/practical_3$ gcc program1.c -o program1
program1.c: In function 'main':
program1.c:10:16: warning: format '%d' expects a matching 'int' argument [-Wformat=]
   10 | printf("PID : %d\n");
      |                ~^
      |                |
      |                int
crazyguy64@CrazyGuy64:~/practical_3$ nano program1.c
crazyguy64@CrazyGuy64:~/practical_3$ nano program1.c
crazyguy64@CrazyGuy64:~/practical_3$ gcc program1.c -o program1
crazyguy64@CrazyGuy64:~/practical_3$ ./program1
Parent process
PID : 1001
Child Process
PID : 1002
PPID : 1001
```

Program 2:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main(){
int i;
for(i=0; i<3;i++){
if(fork()==0){
printf("Child %d PID=%d\n", i+1, getpid());
return 0;
}
}
for(i=0; i<3;i++){
wait(NULL);
}
printf("All children completed.\n");

return 0;
}
```

Output:

```
crazyguy64@CrazyGuy64:~/practical_3$ nano program2.c
crazyguy64@CrazyGuy64:~/practical_3$ gcc program2.c -o program2
crazyguy64@CrazyGuy64:~/practical_3$ ./program2
Child 1 PID=1180
Child 2 PID=1181
Child 3 PID=1182
All children completed.
```